Computer Science & DA

Probability and Statistics

Permutation & Combination

Lecture No. 04

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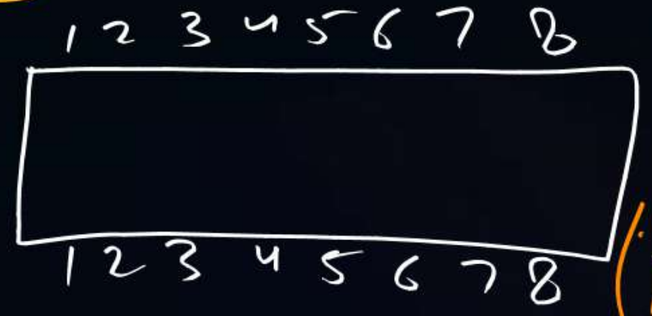
Recap of previous lecture

Topic Permutation and Combination-03

Topics to be Covered

Topic Probability-Basic definition

PQ- 16 persons wants to be seated along a long Table with 8 chairs on each side. Four particular persons wants to be seated on one particular side while two on the other side then find Total number of such arrangements ?



Total arrangements (w/o any restriction) $= 16!$ (Max Ans)

(i) Req. Arrangements $= {}^8P_4 \times {}^8P_2 \times {}^{10}P_{10}$

$= ({}^8C_4 \times 4!) \times ({}^8C_2 \times 2!) \times 10!$

(ii) if side is not particular then Ans $= ? = ({}^2C_1 \times {}^8P_4) \times ({}^1C_1 \times {}^8P_2) \times 10!$

Q: How many 11 letters words can be formed using the letters of the word MATHEMATICS

Sol: MM, AA, TT, E, H, I, C, S $\Rightarrow n = 11$ so Total 11 letter words (RNA) $= \dfrac{11!}{2!\,2!\,2!}$ Ans

Q: How many 4 letter words can be formed using the letters of the word MATHEMATICS

$\underbrace{\text{MM AA TT EHICS}}_{n=11}$ we can make 4 letters word in which there may be

$=$ (All different letters) or (Two alike & Two alike) or (Two alike & Two different)

M, A, T, EHICS MM, AA, TT $\underbrace{\text{Two alike}}_{\Downarrow\ {}^3C_1}$ E, H, I, C, S, A, T $\Downarrow\ {}^7C_2$

$$= \underset{\text{eg (M, T, H, S)}}{\left({}^8C_4 \times 4!\right)} + \underset{\text{eg (MMTT)}}{\left({}^3C_2 \times \frac{4!}{2!\,2!}\right)} + \underset{\text{(MM AH)}}{\left({}^3C_1 \times {}^7C_2 \times \frac{4!}{2!}\right)}$$

MM AA TT EHICS

Explanation of Case III →

Case III: Two alike & Two diff

MM, (A, T, E, H, I, C, S) $\Longrightarrow$ $MM \times {}^7C_2 \times \frac{4!}{2!}$

or

TT, (M, A, E, H, I, C, S) $\Longrightarrow$ $TT \times {}^7C_2 \times \frac{4!}{2!}$

or

AA, (M, T, E, H, I, C, S) $\Longrightarrow$ $AA \times {}^7C_2 \times \frac{4!}{2!}$

Case III → $3 \times {}^7C_2 \times \frac{4!}{2!}$

Q. How many 9 digit Numbers can be formed using the digits 5, 5, 6, 6, 6, 7, 7, 7, 7

Sol: given digits are $n=9$ so 9 digit Numbers $= \dfrac{9!}{2!\,3!\,4!}$

Q. How many 4 digit Numbers (> 6000) can be formed using 5, 5, 6, 6, 6, 7, 7, 7, 7

(a) 54

(b) 27

(c) 81

(d) 51

Total 4 digit Numbers (> 6000) $= \underset{P_1}{(6 \text{ or } 7)}\ \underset{P_2}{(5 \text{ or } 6 \text{ or } 7)}\ \underset{P_3}{(5 \text{ or } 6 \text{ or } 7)}\ \underset{P_4}{(5 \text{ or } 6 \text{ or } 7)}$ (wild guess)

$= 2 \text{ ways} \times 3 \text{ ways} \times 3 \text{ ways} \times 3 \text{ ways}$

$= 54$ Nos (wild Ans)

these 54 Nos include some wrong numbers which are not possible

eg (6555) ✗

eg (6666) ✗

eg (7555) ✗

So Req Ans $= 54 - 3 = 51$

DISTRIBUTION of Different items → (only M. Rule is applicable)

eg In how many ways 10 different Pencils can be distributed among 4 beggars?

(i) if there is No Restriction = ? $= \underset{P_1}{4} \times \underset{P_2}{4} \times \underset{P_3}{4} \times \ldots \times \underset{P_{10}}{4} = 4^{10}$ ways.

(RA)

wrong App $= \underset{B_1}{10} \times \underset{B_2}{10} \times \underset{B_3}{10} \times \underset{B_4}{10} = 10^4$ ✗

(ii) 1st Beggar gets exactly 3 Pencils = ? $= ({}^1C_1 \times {}^{10}C_3 \times 1) \times \underline{3^7}$

(iii) one Beggar gets exactly 3 Pencils $= ({}^4C_1 \times {}^{10}C_3 \times 1) \times \underset{P_1}{3} \times \underset{P_2}{3} \times \underset{P_3}{3} \times \underset{P_4}{3} \times \underset{P_5}{3} \times \underset{P_6}{3} \times \underset{P_7}{3} = {}^4C_1 \times {}^{10}C_3 \times 3^7$

(iv) " " " 3 Particular " $= ({}^4C_1 \times {}^3C_3 \times 1) \times 3^7$

(vi) 1st Beggar gets 3 particular pencils = ? $= ({}^1C_1 \times {}^3C_3 \times 1) \times 3^7 = 3^7$

<u>Distribution of Identical items</u> → n identical items are to be distributed among r persons then possible number of distributions if

(1) Blank groups are allowed (BA) $= {}^{n+r-1}C_{r-1}$ (Formula starts from 0)

(2) Blank groups are not allowed (BNA) $= {}^{n-1}C_{r-1}$ (" " " 1)

eg: In how many ways 10 identical pencils are to be distributed among 4 children if

$n=10$, $r=4$

① Each child gets something = (BNA) = ${}^{n-1}C_{r-1} = {}^{10-1}C_{4-1} = {}^{9}C_3$

② Each child gets at least one pencils = (BNA) = Same

③ " " " one or more Pencils = (BNA) = Same

④ A Child may get nothing = (BA) = ${}^{n+r-1}C_{r-1} = {}^{10+4-1}C_{4-1} = {}^{13}C_3$

⑤ A Child may get any number of pencils = (BA) = Same

Q: 16 Mangoes are to be distributed among 3 beggars then find possible number of distributions if each beggar gets at least 3 Mangoes?

Sol: (M-I) 1st we will distribute 3-3 Mangoes to each beggar & this can be done by 1 way

Now, Remaining Mangoes $= 16-9=7$

& $BA_{(n=7,\, r=3)} = {}^{7+3-1}C_{3-1} = {}^{9}C_{2} = \frac{9\times 8}{2} = 36$

BNA

(M-II) $x+y+z=16$; $(x,y,z)\geq 3$

$x\geq 3,\ y\geq 3,\ z\geq 3$

or $(x-3)\geq 0,\ (y-3)\geq 0,\ (z-3)\geq 0$

$u\geq 0,\ v\geq 0,\ w\geq 0$

$(x-3)+(y-3)+(z-3)=7$

$\boxed{u+v+w=7}$,

$n=7,\ r=3$, $BA = {}^{7+3-1}C_{3-1} = {}^{9}C_{2} = 36$

Various Distributions for $n=7$, $r=3$, BA in $\boxed{u+v+w=7}$

$$\left.\begin{array}{l}(2,2,3)\\(4\ 1\ 2)\\(5\ 2\ 0)\\(1\ 4\ 2)\\ \vdots\\(7\ 0\ 0)\\(0\ 7\ 0)\\(0\ 0\ 7)\end{array}\right\} {}^{9}C_2 = 36 \text{ possibilities}$$

$(0\ 0\ 0) \longrightarrow \times$

In this Quest Blanks are allowed but all Blanks are not allowed just bcoz of SECURITY REASONS

(ii) if BNA then Am = ?

$= {}^{7-1}C_{3-1} = {}^{6}C_2 = 15$

Q. How many non-ve integral solutions are there of $\boxed{x+y+z+w=100}$

Non-ve integers $= 0, 1, 2, 3, 4 \ldots\ldots (W)$

Set of +ve integers $= 1, 2, 3, 4 \ldots\ldots (N)$

$\rightarrow n = 100$ Mangoes, $r = 4$ & BA

So $A_n = {}^{100+4-1}C_{4-1} = {}^{103}C_3$

SET theory

$A = \{a, e, i, o, u\}$ = set of vowels

B = set of good Teachers $= \{ ? \} = \times$

??

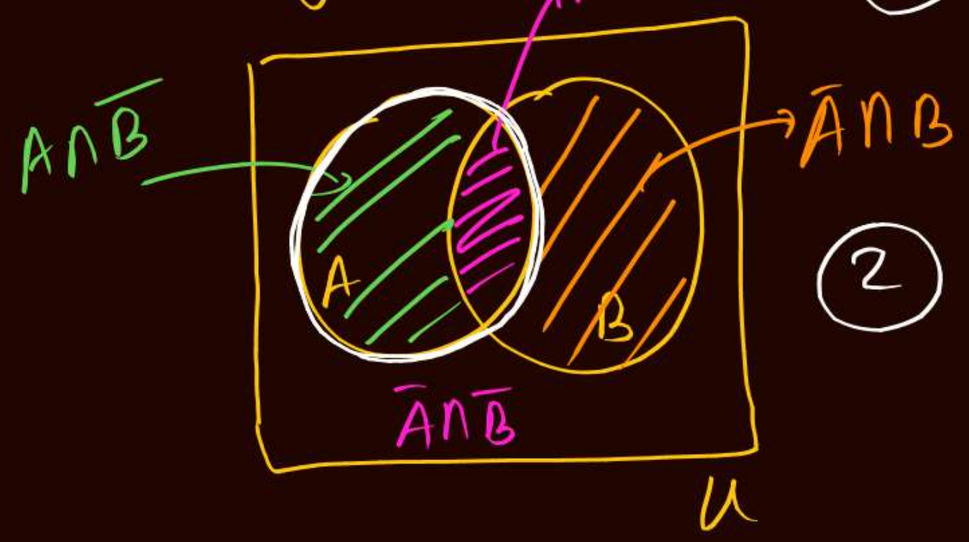
C = set of Teachers having exp $\geq$ 10 yrs $= \{ \checkmark \}$

Def:- " Collection of well defined objects is known as set.

Subset:- $A_1 = \{a, e\}$, $A_2 = \{e, i, u\}$, $A_3 = \{i, o, u, a\}$ - - - - - -

Null set $\phi = \{\ \}$ is also subset & A is also subset of A

(4) VENN Diag Appl-e



$$\boxed{A-B = A - A\cap B}$$

only $A = A - A\cap B$

(1) $\underbrace{(A\cap\bar{B}) \cup (\bar{A}\cap B) \cup (A\cap B)}_{A\cup B} \cup \underbrace{(\bar{A}\cap\bar{B})} = U$

(2) $A\cap B \Rightarrow$ A & B

$\bar{A}\cap\bar{B} \Rightarrow$ Neither A Nor B

$A\cup B \Rightarrow$ Either A or B / At least one of A or B or Both

$A\cap\bar{B} \Rightarrow$ only $A = A - A\cap B$

$\bar{A}\cap B \Rightarrow$ only $B = B - A\cap B$

$$\boxed{(\text{Either A or B}) + (\text{Neither A Nor B}) = \text{Total}}$$

i.e (1) $\boxed{\text{Neither A Nor B} = \text{Total} - \text{Either A or B}}$

(2) Either A or B = Total − Neither A Nor.

$\boxed{(\text{At least one of A or B or Both}) = \text{Total} - \text{None}}$

Q. out of 1-1000 Natural Numbers (both inclusive), how many of them are ?

(1) Neither divisible by 3 Nor by 5 Nor by 7

(2) Divisible by 3 & 5 but not by 7

$n(3) = 1000/3 = 333$

$n(5) = 1000/5 = 200$

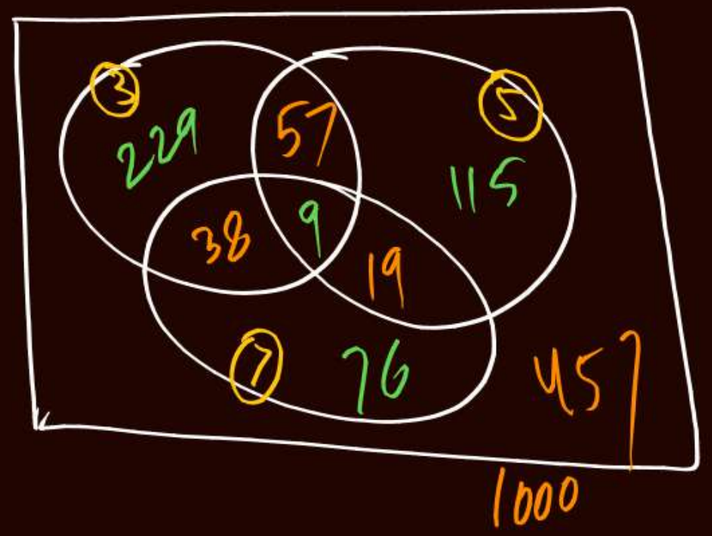
$n(7) = 1000/7 = 142$

$n(3 \cap 5) = n(15) = 1000/15 = 66$

$n(5 \cap 7) = n(35) = 1000/35 = 28$

$n(3 \cap 7) = n(21) = 1000/21 = 47$

$n(3 \cap 5 \cap 7) = n(105) = 1000/105 = 9$



$n(\text{only } 3) = 333 - (57 + 38 + 9) = 229$ Ans

$n(\text{only } 5) = 200 - (57 + 9 + 19) = 115$ Ans

$n(\text{only } 7) = 142 - (38 + 9 + 19) = 76$ Ans

$n(3 \text{ or } 5 \text{ or } 7) = n(3 \cup 5 \cup 7) = ?$

$= (229 + 115 + 76) + (57 + 38 + 19) + (9) = 543$

$n(\overline{3} \cap \overline{5} \cap \overline{7}) = \text{Total} - n(3 \cup 5 \cup 7)$

$= 1000 - 543 = 457$

$n(3 \cap 5 \cap \overline{7}) = ? = 57$

$n(\text{divisible by exactly 2 of them}) = ? = 57 + 38 + 19$

$= 114$

THANK - YOU